

In C#, you can use the **System.IO** namespace for file handling, directory creation, and deletion.

Here's a basic example that demonstrates reading from and writing to a file, as well as creating and deleting directories.

1.Namespace and Class Declaration:

```
namespace File_handling_Directory_creation_deletion
{
    class Program
    {
        static void Main(string[] args)
        {
        }
    }
}
```

- The code starts with a namespace declaration named `File_handling_Directory_creation_deletion`.
- Inside the namespace, there's a class named `Program` with a `Main` method, which serves as the entry point of the program.

2.Directory Creation:

```
// Specify the path to a directory
string directoryPath = @"C:\ExampleDirectory";

// Check if the directory exists, if not, create it
if (!Directory.Exists(directoryPath))
{
    Directory.CreateDirectory(directoryPath);
    Console.WriteLine("Directory created successfully.");
}
else
{
    Console.WriteLine("Directory already exists.");
}
```

- Specifies a directory path (directoryPath).
- Checks if the directory exists using Directory.Exists.
- If the directory doesn't exist, it is created using Directory.CreateDirectory.
- A message is printed to the console indicating whether the directory was created or already exists.

3.File Handling:

```
// Specify the path to a file within the directory
string filePath = Path.Combine(directoryPath, "example.txt");
```

```
// Write to a file
WriteToFile(filePath, "Hello, this is a test message.");
```

```
// Read from a file
string content = ReadFromFile(filePath);
Console.WriteLine($"File content: {content}");
```

- Specifies a file path (filePath) by combining the directory path with the file name ("example.txt") using Path.Combine.
- Writes a test message to the file using the WriteToFile method.
- Reads the content of the file using the ReadFromFile method and prints it to the console.

4.File Writing and Reading Methods:

```
static void WriteToFile(string filePath, string content)
{
    try
    {
        // Open the file for writing
        using (StreamWriter writer = new StreamWriter(filePath))
        {
            // Write content to the file
            writer.Write(content);
            Console.WriteLine("File written successfully.");
        }
    }
    catch (Exception ex)
    {
        Console.WriteLine($"Error writing to file: {ex.Message}");
    }
}
```

```
}
```

```
static string ReadFromFile(string filePath)
```

```
{
```

```
    try
```

```
    {
```

```
        // Open the file for reading
```

```
        using (StreamReader reader = new StreamReader(filePath))
```

```
        {
```

```
            // Read content from the file
```

```
            return reader.ReadToEnd();
```

```
        }
```

```
    }
```

```
    catch (Exception ex)
```

```
    {
```

```
        Console.WriteLine($"Error reading from file: {ex.Message}");
```

```
        return string.Empty;
```

```
    }
```

```
}
```

- WriteToFile method opens a file for writing using StreamWriter and writes the specified content to the file.
- ReadFromFile method opens a file for reading using StreamReader and reads the content from the file.

5.File Deletion:

```
// Delete the file
```

```
DeleteFile(filePath);
```

- Deletes the file specified by filePath using the DeleteFile method.

6.File and Directory Deletion Methods:

```
static void DeleteFile(string filePath)
{
    try
    {
        // Check if the file exists before attempting to delete
        if (File.Exists(filePath))
        {
            File.Delete(filePath);
            Console.WriteLine("File deleted successfully.");
        }
        else
        {
            Console.WriteLine("File does not exist.");
        }
    }
    catch (Exception ex)
    {
        Console.WriteLine($"Error deleting file: {ex.Message}");
    }
}
```

```
static void DeleteDirectory(string directoryPath)
{
    try
    {
        // Check if the directory exists before attempting to delete
        if (Directory.Exists(directoryPath))
        {
            Directory.Delete(directoryPath);
            Console.WriteLine("Directory deleted successfully.");
        }
        else
        {
            Console.WriteLine("Directory does not exist.");
        }
    }
}
```

```
catch (Exception ex)
{
    Console.WriteLine($"Error deleting directory: {ex.Message}");
}
}
```

- **DeleteFile** method checks if the file exists using **File.Exists** before attempting to delete it with **File.Delete**.
- **DeleteDirectory** method checks if the directory exists using **Directory.Exists** before attempting to delete it with **Directory.Delete**.

7. Console Exit Message:

```
Console.WriteLine("Press any key to exit.");
Console.ReadKey();
```

Displays a message indicating to press any key to exit the program.
Waits for a key press using Console.ReadKey().

This program demonstrates basic file handling and directory creation/deletion operations in C#.

Example

Part 1: //DirectoryAndFileHandler.cs

```
using System;
using System.IO;

public class DirectoryAndFileHandler
{
    // Method to create a directory
    public void CreateDirectory(string directoryPath)
    {
        if (!Directory.Exists(directoryPath))
        {
            Directory.CreateDirectory(directoryPath);
            Console.WriteLine("Directory created: " + directoryPath);
        }
        else
        {
            Console.WriteLine("Directory already exists: " + directoryPath);
        }
    }

    // Method to create and write to a file
    public void WriteToFile(string filePath, string content)
    {
        File.WriteAllText(filePath, content);
        Console.WriteLine("File created and content written: " + filePath);
    }

    // Method to read from a file
    public string ReadFromFile(string filePath)
    {
        if (File.Exists(filePath))
        {
            return File.ReadAllText(filePath);
        }
        else
        {
            return null;
        }
    }
}
```

```

        Console.WriteLine("File not found: " + filePath);
        return string.Empty;
    }
}

// Method to append content to a file
public void AppendToFile(string filePath, string content)
{
    if (File.Exists(filePath))
    {
        File.AppendAllText(filePath, content);
        Console.WriteLine("Content appended to file: " + filePath);
    }
    else
    {
        Console.WriteLine("File not found: " + filePath);
    }
}

// Method to delete a file
public void DeleteFile(string filePath)
{
    if (File.Exists(filePath))
    {
        File.Delete(filePath);
        Console.WriteLine("File deleted: " + filePath);
    }
    else
    {
        Console.WriteLine("File not found: " + filePath);
    }
}

```

```

// Method to delete a directory
public void DeleteDirectory(string directoryPath)
{
    if (Directory.Exists(directoryPath))
    {
        Directory.Delete(directoryPath, true); // true to delete subdirectories and files
        Console.WriteLine("Directory deleted: " + directoryPath);
    }
    else
    {
        Console.WriteLine("Directory not found: " + directoryPath);
    }
}
}

```

Part 2://Program.cs (Main Program)

```
using System;
```

```

class Program
{
    static void Main()
    {
        // Specify the desired location
        string specificLocation = @"C:\MyCustomDirectory";
        string directoryPath = System.IO.Path.Combine(specificLocation, "ExampleDirectory");
        string filePath = System.IO.Path.Combine(directoryPath, "example.txt");

        // Create an instance of the handler class
        DirectoryAndFileHandler handler = new DirectoryAndFileHandler();

        // Perform operations
        handler.CreateDirectory(directoryPath);
        handler.WriteToFile(filePath, "Hello, this is a sample text file in a specific location.");
        Console.WriteLine("Content of the file:");
        Console.WriteLine(handler.ReadFromFile(filePath));

        handler.AppendToFile(filePath, "\nThis is an appended line.");
        Console.WriteLine("Updated content of the file:");
    }
}

```

```
        Console.WriteLine(handler.ReadFromFile(filePath));

        handler.DeleteFile(filePath);
        handler.DeleteDirectory(directoryPath);
    }
}
```

Sample Output

Directory created: C:\MyCustomDirectory\ExampleDirectory

File created and content written: C:\MyCustomDirectory\ExampleDirectory\example.txt

Content of the file:

Hello, this is a sample text file in a specific location.

Content appended to file: C:\MyCustomDirectory\ExampleDirectory\example.txt

Updated content of the file:

Hello, this is a sample text file in a specific location.

This is an appended line.

File deleted: C:\MyCustomDirectory\ExampleDirectory\example.txt

Directory deleted: C:\MyCustomDirectory\ExampleDirectory